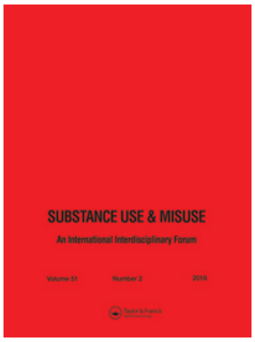

Substance Use & Misuse

ISSN: 1082-6084 (Print) 1532-2491 (Online) Journal homepage: https://www.tandfonline.com/loi/isum20

High Prevalence and Incidence of HIV and HCV Among New Injecting Drug Users With a Large Proportion of Migrants—Is Prevention Failing?

Cinta Folch, Jordi Casabona, Albert Espelt, Xavier Majó, Mercè Meroño, Victoria Gonzalez, Lucas Wiessing, Joan Colom, M. Teresa Brugal & REDAN Study Group

SUBSTANCE USE & MISUSE
2016, VOL. 51, NO. 2, 250–260
http://dx.doi.org/10.3109/10826084.2015.1092991

Taylor & Francis
Taylor & Francis Group

ORIGINAL ARTICLE

High Prevalence and Incidence of HIV and HCV Among New Injecting Drug Users With a Large Proportion of Migrants—Is Prevention Failing?

Cinta Folch[a,b,c], Jordi Casabona[a,b,c,d], Albert Espelt[b,e,f], Xavier Majó[g], Mercè Meroño[h], Victoria Gonzalez[a,b,i], Lucas Wiessing[j], Joan Colom[g], M. Teresa Brugal[b,e], and REDAN Study Group

[a]Centre d'Estudis Epidemiològics sobre les Infeccions de Transmissió Sexual i Sida de Catalunya (CEEISCAT), Agéncia de Salut Pública de Catalunya (ASPC), Generalitat de Catalunya, Badalona, Spain; [b]CIBER Epidemiología y Salud Pública (CIBERESP), Spain; [c]Fundació Institut d'Investigació Germans Trias i Pujol (IGTP), Badalona, Spain; [d]Departament de Pediatria, d'Obstetrícia i Ginecologia i de Medicina Preventiva i de Salut Pública, Facultat de Medicina, Universitat Autònoma de Barcelona, Bellaterra (Cerdanyola del Vallés), Spain; [e]Agéncia de Salut Pública de Barcelona, Spain; [f]Departament de Psicologia i Metodologia de les Ciéncies de la Salut, Universitat Autònoma de Barcelona, Bellaterra (Cerdanyola del Vallés), Spain; [g]Subdirecció General de Drogodependéncies, Agéncia Salut Pública de Catalunya (ASPC), Departament de Salut de la Generalitat de Catalunya; [h]Ámbit Prevenció, Barcelona, Spain; [i]Microbiology Service, Hospital Universitari Germans Trias i Pujol, Badalona, Spain; [j]European Monitoring Centre for Drugs and Drug Addiction (EMCDDA), Lisbon, Portugal

ABSTRACT
Objectives: The aim of this study was to assess differences in the prevalence of HIV and HCV infection and associated risk factors between new (injecting for ≤5 years) and long-term injectors and to estimate HIV/HCV incidence among new injectors. *Methods:* Cross-sectional study among people who inject drugs (PWID) who attended harm reduction centers in Catalonia in 2010–11. Anonymous questionnaires and oral fluid samples were collected. Poisson regression models were applied to determine the association between HIV/HCV infection and risk factors. *Results:* Of the 761 participants, 21.4% were new injectors. New injectors were younger than long-term injectors (mean age = 31.6 vs. 37.8) and were more likely to be immigrants (59.0% vs. 33.4%). HIV and HCV prevalence was 20.6% and 59.4% among new injectors, and estimated HIV and HCV incidence 8.7 and 25.1 /100 person-years, respectively. Among new injectors, HIV infection was associated with homelessness (PR = 3.10) and reporting a previous sexually transmitted infection (PR = 1.79). Reporting front/backloading (PR = 1.33) and daily injection (PR = 1.35) were risk-factors for HCV infection. For long-term injectors, HIV risk factors were: having shared syringes (PR = 1.85), having injected cocaine (PR = 1.38), reporting front/backloading (PR = 1.30) and ever having been in prison (PR = 2.03). *Conclusion:* A large proportion of PWID in Catalonia are new injectors, a subgroup with a high level of both sexual and parenteral exposure and a high incidence rate of HIV/ HCV infections. It is important to improve early diagnosis of these infections among this group, in particular among migrants. To identify and address risk factors for homelessness PWID should be a priority.

KEYWORDS
Hepatitis C; HIV, injecting drug use; migrants; new injectors

Introduction

People who inject drugs (PWID) are at high risk of acquiring various blood-borne pathogens such as human immunodeficiency virus (HIV) and hepatitis C virus (HCV). While most new injectors do not become infected with HIV during the first few years, most HCV infections occur within the first 6 years of injecting drug use, and prevalence can exceed 50% after 1 year of injecting (Garten et al., 2004; Judd et al., 2005; Neaigus et al., 2007; Hagan, Pouget, Des Jarlais, & Lelutiu-Weinberger, 2008). New injectors are thus a key population for monitoring blood-borne infections as they reflect recent infections as well as may forebode changes in the epidemiology of injecting drug use.

Although the increasing availability of harm reduction programs in Catalonia (Barrio et al., 2012) has contributed to reductions in the prevalence of HIV among PWID (de la Fuente et al., 2006; EMCDDA, 2012), the prevalence of HCV remains high and exceeds 75%-80% (Huntington, Folch, Gonzalez, Meroño, Ncube, & Casabona, 2010). Data from samples of young injectors indicate ongoing transmission of HIV and HCV (Bravo et al., 2012) and are similar to those reported in Estonia, France, and Poland (Wiessing, Likatavicius, Hedrich, Guarita, van de Laar, & Vicente, 2011) and consistent with the persistent high levels of risk behaviors in this population (Hacker et al., 2005). Key risk factors for HIV and HCV infection among new injectors include high frequency of injecting/sharing, networking with high-risk

CONTACT Cinta Folch cfolch@iconcologia.net Centre for Epidemiological Studies on HIV/STI in Catalonia (CEEISCAT), Hospital Universitari Germans Trias i Pujol, Ctra de Canyet s/n, 08916 Badalona, Spain.

injectors, and low HIV/HCV risk perception (Miller et al., 2002; Hacker et al., 2005).

Factors associated with HIV and HCV infection among PWID have been widely studied in Spain as in other European countries, however very few studies have looked at the factors associated among the population of new injectors. To assess the profile of new injectors in comparison with long-term injectors may help to understand the recent dynamics on HIV and HCV in this population and to implement appropriate prevention strategies. Moreover, as the number of migrant injectors has increased considerably in Catalonia in recent years (Tordable Merino, Sánchez Sánchez, Santos Sanz, García Vicario, & Redondo Martín, 2010; Freixa & Meroño, 2011) and previous studies reported important differences in comparison with native injectors (Saigí et al., 2014), by considering migrant population among this analysis preventive strategies can be more easily adapted to the reality of drug consumption in our setting.

The objective of this study was to assess potential differences in the prevalence of HIV and HCV infection as well as their associated risk factors between new and long-term injectors in Catalonia. Moreover, incidence of HIV and HCV infection were estimated among the group of new injectors.

Methods

Design and participants

A cross-sectional study was carried out during 2010–2011 in the network of harm reduction centres as part of the Integrated HIV and Sexually Transmitted Infections (STI) Surveillance System in Catalonia (SIVES) (CEEIS-CAT, 2013). These centres provide needle exchange programs, outreach programs, and supervised injecting facilities. After collecting data on the number and characteristics of clients contacted by these centres in the previous year, a stratified convenience sample of PWID was selected according to the type of centre and country of origin using proportional allocation.

Individuals who reported having injected in the previous 6 months and who attended these centres were eligible to take part. Those who agreed to participate signed the informed consent document and received €24.

Data collection

Face-to-face interviews were conducted at the harm reduction centers by trained interviewers using an anonymous structured questionnaire adapted from the WHO Multy-city study on drug injecting and risk of HIV infection (WHO, 1994). The questionnaire took approximately 35 minutes to complete and was translated into Spanish, Romanian, Russian, English, and French and included questions on sociodemographic characteristics (country of origin, age, sex, education level, occupational status, place of residence, treatment for drug addiction), drug use (time since first injection, age at first injection, frequency of injection, sharing of syringes and/or other injecting paraphernalia such as water containers, spoons and filters, drugs used), sexual practices (commercial sex work, use of condoms with steady and casual partners), knowledge of HIV and HCV status, and previous history of STI. Most questions on behavior referred to the preceding 6 months. For the purpose of the analysis, "new" injectors were defined as those who had been injecting for 5 years or less, while "long-term" injectors were those who had been injecting for more than 5 years. In response to the question, "What is your country of origin?, individuals who did not indicate that they were from Spain were classified as immigrants. Undiagnosed HIV and/or HCV infection was defined as a positive biological test result from any participant self-reporting as "negative" or who had never been tested.

Oral fluid samples were also taken anonymously to determine the prevalence of HIV and HCV infection. Anti-HIV antibodies were detected in oral fluid using Genscreen HIV-1/2 Version 2.0 assay from Bio-Rad (sensitivity = 98.5%; specificity = 100%) (Chohan et al., 2001); anti-HCV antibodies were detected using HCV 3.0 SAVe ELISA (sensitivity = 86.7%; specificity = 100%) (Judd et al., 2003). Questionnaires and oral fluid samples were linked using an anonymous numerical code.

Statistical analysis

Proportions were compared between new and long-term injectors using the Pearson χ^2 and the Fisher exact tests. For quantitative variables, means were compared using the *t* test after verification of the equality of variances using the Levene test.

HIV and HCV incidence among new injectors was estimated using the following assumptions: (1) all of them were HIV and HCV seronegative when they began injecting; (2) they became infected at the midpoint between beginning to inject and the time of oral fluid sample collection; (3) no HIV seropositives were lost to AIDS or for other reasons among the new injectors. The time at risk for HIV/HCV seronegative new injectors was the total time from first injection to the time of the interview. The estimated HIV and HCV incidence rate was the number of HIV and HCV seropositive new injectors divided by the sum of the time at risk for the HIV and HCV seropositive new injectors and the time at risk for the HIV seronegative new injectors (Uusküla et al., 2008).

Univariate and Poisson regression models with robust variance were applied to determine the association between HIV and HCV infection and sociodemographic and behavioral characteristics among new and long-term injectors (Barros & Hirakata, 2003). Variables with a significance level <0.10 in the univariate analysis were included in the multivariate analysis, after adjusting for age, gender, years of injection and country of origin, and the Prevalence Ratio (PR) with its respective 95%CI was calculated. Final multivariate models were derived using stepwise backward elimination process. Statistical significance was set at $P < 0.05$. The analyses were performed using SPSS version 17.0.

Results

Of the 761 PWID enrolled in the study, 21.4% reported injecting drugs for 5 years or less before the interview. Table 1 shows the main sociodemographic characteristics, sexual practices, and drug use behaviors of the participants by duration of injection.

New injectors were younger than long-term injectors (mean age = 31.6 vs. 37.8) and reported a higher level of education (16.8% vs. 25.2% primary education or less). In addition, they were more likely to be immigrants (59.0% vs. 33.4%), mainly from Eastern European countries (62.1% and 54.5%, respectively). More long-term injectors had ever been in prison (71.5% vs. 49.7%) and were currently in treatment for drug use (57.8% vs. 40.4%). Fewer new injectors had begun injecting at very young ages (3.7% vs. 42.5% <18 years) or reported ever sharing syringes (33.5% vs. 56.1%), although the proportion who reported daily injection was higher among new injectors (55.3% vs. 44.7%) (Table 1).

Regarding injecting and sexual risk practices during the last 6 months, no differences were found between new and long-term injectors except that the prevalence of sharing injecting paraphernalia other than syringes and needles during the last 6 months was higher among new injectors (55.9% vs. 47.2%). The lifetime prevalence of self-reported STI was similar between long-term (14.0%) and new injectors (17.8%). The proportion of ever tested for HIV was lower among new injectors (90.7% and 95.9%, respectively), as well as the proportion of ever tested for HCV (83.0% and 96.4%, respectively) (Table 1).

The prevalence of HIV and HCV infection in the total sample was 33.2% (95%CI, 29.8–36.5) and 72.0% (95%CI, 68.8–75.2), respectively. Figure 1 shows the prevalence of HIV and HCV infection according to injection history. HCV prevalence is significantly associated with time since first injection, increasing from 59.4% (95%CI, 51.8–67.0) in the new injectors increasing to 77.1% (95%CI, 73.4–80.9) in those with an injection history of more than 10 years. The prevalence of HIV was already high among new injectors (20.6%; 95%CI, 14.4–56.9) and those with an injection history of 5–10 years (20.4%; 95%CI, 12.9–27.8) while even much higher among participants with an injection history of more than 10 years (40.5%; 95%CI, 36.1–44.9).

The estimated HIV incidence among new injectors was 8.71/100 person-years at risk, and the HCV incidence was 25.06/100 person-years.

Table 1. Sociodemographic, sexual, and drug use behaviors of new and long-term injectors.

	New $N = 161$%	Long-term $N = 593$%	p
Median age (SD)	31.6 (7.5)	37.8 (7.1)	<.0001
Immigrants	59.0	33.4	<.0001
Female	21.1	16.2	NS
Education level: primary or less	16.8	25.2	.026
Homeless[1]	22.4	16.9	NS
Living alone[1]	31.1	29.8	NS
Receiving a retirement pension/incapacity benefit/social fund	15.5	34.9	<.0001
Main source of income: illegal	55.7	45.9	.029
Ever in prison	49.7	71.5	<.0001
Currently in treatment (to reduce or stop injecting)	40.4	57.8	<.0001
Age at first injection: < 18 years	3.7	42.5	<.0001
Injection of cocaine[1]	55.9	65.7	.022
Injection of heroin[1]	80.7	78.5	NS
Injection of speedball[1]	54.7	56.3	NS
Frequency of injection: daily[1]	55.3	44.7	.017
Ever sharing syringes	33.5	56.1	<.0001
Accepted used syringes[1]	14.3	12.0	NS
Passing on used syringes[1]	17.4	13.5	NS
Front/backloading[1]	13.9	12.5	NS
Sharing other injecting equipment[1]	55.9	47.2	.05
Steady sexual partner[1]	48.4	48.6	NS
Unprotected sexual intercourse (steady partner)[1,2]	80.8	69.9	NS
IDU steady sexual partner[1,2]	43.6	42.0	NS
HIV-positive steady sexual partner[1,2]	10.3	18.5	NS
Casual sexual partner[1]	39.8	34.4	NS
Unprotected sexual intercourse (casual partner)[1,3]	39.7	32.7	NS
Commercial sex[1]	7.5	4.7	NS
Men who had sex with men (last 5 years)	4.3	3.5	NS
History of sexually transmitted infections	17.8	14.0	NS
Ever tested for HIV	90.7	95.9	.008
Ever tested for HCV	83.0	96.4	<.0001

[1]Last 6 months
[2]Among those with steady partners
[3]Among those with casual partners
[4]NS = Nonsignificant

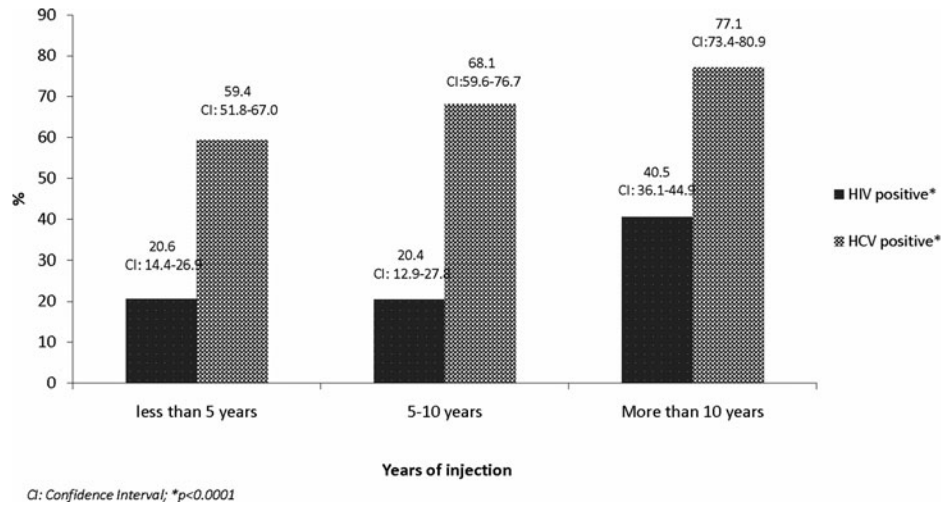

Figure 1. HIV and HCV prevalence by duration of injection.

More new injectors than long-term injectors were unaware of their HIV infection (48.5% vs. 20.3%; $p < .0001$) and of their HCV positive antibody status at the time of the interview (50.5% vs. 20.2%; $p < .0001$). Among new injectors, unawareness of HIV and HCV infection was higher among migrants (71.4% and 57.4%, respectively), although differences were not statistically significant. Among long-term injectors, the proportion of unaware HIV and HCV infections among migrants (43.2% and 25.6%, respectively) were higher than among native ($p < .05$) (Figure 2).

The univariate results of the variables associated with HIV and HCV seroprevalence are shown in Table 2. Among both new and long-term injectors, risk factors for HIV infection were being born in Spain, ever having shared syringes, having injected cocaine, having accepted used syringes and reporting front/backloading during the last 6 months, ever having been in prison and reporting a previous STI. Additional factors associated with HIV infection among new injectors were reporting an illegal source of income, being homeless, injection of speedball and commercial sex in the last 6 months. No common HCV risk factors for both new and long-term injectors emerged at univariate level. Among new injectors, HCV infection was associated with being a migrant, ever having shared syringes, reporting daily injection and front/backloading during the last 6 months, and reporting a previous STI. Ever having shared syringes was almost significant as the only risk factor for HCV among long-term injectors.

In the adjusted multivariate models, risk factors for HIV infection among new injectors were being homeless (PR = 3.10) and reporting a previous STI (PR = 1.79). In contrast, having reported unprotected sex with steady partners during the last 6 months was predictive for being HIV-negative (PR = 0.42). On the other hand, reporting front/backloading in the last 6 months (PR = 1.33) and daily injection (PR = 1.35) were risk factors for HCV infection in the multivariate analysis for new injectors (Table 3).

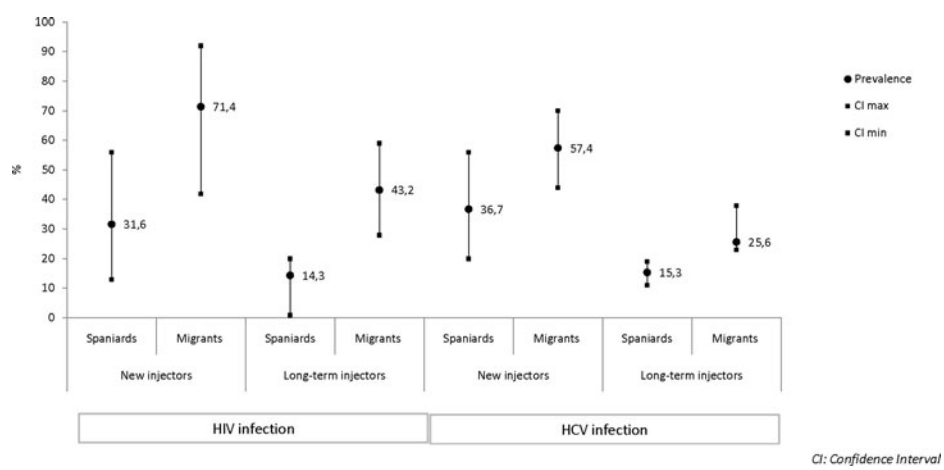

Figure 2. Unknown HIV and HCV infections among new and long-term injectors by country of origin.

Table 2. Univariate associations between sociodemographic characteristics, sexual and drug use behaviors, and self-reported sexually transmitted infections for new and long-term injectors.

	New injectors						Long-term injectors					
	HIV-positive			HCV-positive			HIV-positive			HCV-positive		
	PR	95% CI	*p*	PR	95% CI	*p*	PR	95% CI	*p*	PR	95% CI	*p*
Born in Spain	1.93	1.05–3.57	.036	0.69	0.52–0.92	.012	1.84	1.39–2.42	<.0001	1.03	0.93–1.14	.571
More than 30 years old	1.26	0.68–2.33	.464	1.12	0.87–1.45	.384	2.51	1.56–4.03	<.0001	1.16	0.99–1.34	.061
Female	1.61	0.85–3.05	.143	1.05	0.78–1.42	.744	1.17	0.90–1.52	.246	0.93	0.81–1.06	.288
In treatment (last 12 months)	1.08	0.58–1.99	.813	0.93	0.71–1.22	.605	1.22	0.97–1.52	.083	1.06	0.97–1.17	.195
Primary education or less	0.39	0.16–0.92	.032	0.70	0.48–1.03	.069	0.83	0.61–1.13	.237	1.06	0.90–1.25	.489
Main source of income: illegal[1]	2.51	1.21–5.22	.013	1.09	0.84–1.41	.535	0.90	0.72–1.12	.348	0.93	0.84–1.02	.115
Homeless[1]	3.36	1.90–5.95	<.0001	1.21	0.92–1.58	.174	1.08	0.82–1.42	.581	0.85	0.73–0.99	.036
Injection of cocaine[1]	2.13	1.06–4.28	.034	0.89	0.69–1.14	.354	1.49	1.16–1.92	.002	0.92	0.84–1.01	.090
Injection of heroin[1]	0.75	0.38–1.50	.418	0.96	0.70–1.32	.806	0.84	0.66–1.07	.170	0.90	0.82–0.99	.038
Injection of speedball[1]	2.24	1.11–4.51	.024	1.26	0.96–1.64	.092	1.17	0.94–1.46	.148	1.00	0.91–1.09	.960
Daily injection[1]	1.88	0.96–3.69	.066	1.40	1.06–1.85	.017	0.93	0.75–1.16	.531	1.06	0.97–1.17	.182
Age at first injection <18 years	0.80	0.13–4.93	.812	1.43	0.97–2.09	.069	1.50	1.21–1.85	<.0001	1.03	0.94–1.13	.569
Having shared syringes	2.09	1.15–3.80	.016	1.31	1.02–1.68	.033	2.41	1.85–3.14	<.0001	1.10	1.00–1.21	.053
Accepted used syringes[1]	2.23	1.19–4.18	.012	1.12	0.80–1.55	.512	1.64	1.29–2.09	<.0001	1.07	0.94–1.21	.311
Sharing other injecting equipment[1]	1.83	0.93–3.60	.077	1.20	0.92–1.56	.187	1.01	0.82–1.25	.921	0.94	0.86–1.04	.220
Front/backloading[1]	3.07	1.74–5.41	<.0001	1.47	1.15–1.89	.002	1.56	1.21–2.00	.001	1.01	0.88–1.16	.866
Unprotected sex (steady partner)[1]	0.41	0.19–0.90	.025	0.94	0.72–1.23	.647	0.57	0.44–0.75	<.0001	0.97	0.88–1.07	.560
Unprotected sex (casual partner)[1]	0.35	0.09–1.35	.127	0.70	0.44–1.11	.132	0.63	0.41–0.98	.042	1.07	0.94–1.22	.275
Commercial sex[1]	2.74	1.42–5.31	.003	1.29	0.91–1.84	.158	0.98	0.59–1.63	.930	0.89	0.69–1.16	.395
Having been in prison	2.36	1.20–4.63	.013	1.29	1.00–1.68	.053	2.38	1.70–3.32	<.0001	1.08	0.97–1.20	.175
Sexually transmitted infections (ever)	2.29	1.26–4.15	.007	1.31	1.01–1.71	.040	1.60	1.27–2.03	<.0001	0.97	0.84–1.11	.635

[1] Last 6 months; CI: Confidence Interval; PR: Prevalence Ratio.

Table 3. Risk factors for HIV and HCV infections. Poisson regression models*.

	PR	95% CI	*p*
New injectors			
HIV infection			
Homeless[1]	3.10	1.80–5.33	<.001
Unprotected sex (steady partner)[1]	0.42	0.21–0.82	.012
Sexually transmitted infections (ever)	1.79	1.10–2.90	.018
HCV infection			
Front–backloading[1]	1.33	1.01–1.76	.043
Daily injection[1]	1.35	1.20–1.79	.036
Long-term injectors			
HIV infection			
Injection of cocaine[1]	1.38	1.09–1.76	.009
Having shared syringes (ever)	1.85	1.43–2.40	<.001
Frontbackloading[1]	1.30	1.06–1.60	.012
In prison (ever)	2.03	1.45–2.85	<.001
Unprotected sex (steady partner)[1]	0.63	0.49–0.82	<.001
Unprotected sex (casual partner)[1]	0.62	0.40–0.94	.025
HCV infection			
Homeless[1]	0.84	0.72–0.97	<.001

[1] Last 6 months; *After adjusting for age, country of origin, gender and years of injection
PR, prevalence ratio; CI: Confidence Interval

For long-term injectors (Table 3), risk factors for HIV at multivariate level were ever having shared syringes (PR = 1.85), having injected cocaine (PR = 1.38) and reporting front/backloading (PR = 1.30) during the last 6 months and ever having been in prison (PR = 2.03). Being homeless was negatively associated with HCV infection among long-term injectors (PR = 0.84).

Discussion

The high HIV and HCV prevalence found in this study among PWID within the first 5 years of injecting in Catalonia emphasizes the importance of strengthening early prevention interventions targeting new injectors. The estimated HCV incidence based on prevalence data among this group (25.06/100 person-years) confirms that most new infections occurred very soon after initiation of injection, consistent with previous studies (Maher, Li, Jalaludin, Chant, & Kaldor, 2007; Neaigus et al., 2007). This is similar to previous findings in Spain among new injectors younger than 30 years old (Bravo et al., 2012). Although the estimated HIV incidence indicates ongoing and high levels of HIV transmission among new injectors (8.71/100 person-years), it is confirmed that HIV does not spread as efficiently as HCV in this group.

Migrants represent a higher proportion of the new IDUs population in Catalonia, as compared to long-term IDUs, suggesting an increased influx of migrants among this population as reported over the last years in Athens (Paraskevis et al., 2013, Sypsa et al., 2015), but also a decrease in the autochthonous population who start using injected drugs. These findings highlight the need for further research on the influence of migration on drug using patterns and determinants of the behaviors so that drug services can remain responsive to their needs. When injection drug use and mobility overlap, risk for HIV transmission to other populations and geographic areas increases (Rachlis et al., 2007; Strathdee et al., 2008; Friedman, Rossi, Braine, 2009) suggesting that policies in Europe need to be multifaceted, recognizing the specific needs of migrant populations and working to increase their access to health services and if possible prevent their transition to injection drug use.

Although a higher percentage of long-term injectors had shared needles at some time during their life, probably due to the effect of cumulative risk over time, no differences emerged between the groups when asked about risk behaviors in the last 6 months, except for the practice of indirect sharing, in which prevalence was higher among new injectors. Earlier studies have suggested a higher frequency of risk behaviors for HIV and HCV infection among new injectors than among more experienced users (Becker Buxton et al., 2004; Hacker et al., 2005).

Factors associated with young age itself, such as impulsivity and lower perception of risk may contribute to the elevated levels of risk practices among new injectors (Bailey et al., 2007). On the other hand, differences between the contextual environments of new and long-term injectors (e.g., inject in public spaces, fewer socioeconomic resources, unstable housing) may contribute to risky behaviors by recent initiates into injection (Paraskevis et al., 2013). Future studies in Catalonia will be necessary to better characterize some of the behavioral characteristics of new injectors.

Multivariate analysis revealed differences in the risk of HIV and/or HCV infection among new and long-term injectors. Firstly, the risk of HCV infection among new injectors was higher in those who had practiced front/backloading during the previous 6 months. Although many injectors recognized the risk of viral transmission associated with sharing needles and syringes, there is relatively little understanding of how HIV/HCV transmission is associated with sharing injection paraphernalia (Thiede et al., 2007). This explains in part why, while syringe sharing has declined among PWID since the introduction of harm-reduction programs, shared use of drug preparation equipment has persisted (De la Fuente et al., 2006; Hagan et al., 2010). The association between practices of sharing injection equipment indirectly and these infections has also been reported previously at national and international level (Quan et al., 2009; Pouget, Hagan, & Des Jarlais, 2012). Although indirect sharing was also associated with HIV infection among long-term injectors, the main risk factors for HIV among this group were having ever shared syringes and having ever been in prison, both time dependent variables which reflect an effect of cumulative exposure. The delay in the implementation of Needle Exchange Programs in Catalonia, in particular in prisons, in which it was not fully implemented until 2006, could also explain in part the association of being in prison a HIV infection among this population.

Homelessness facilitates the acquisition of risk behaviors and differences in health status among injectors (Reyes et al., 2005). In fact, numerous studies have shown high HIV prevalence among homeless injectors (Magura, Nwakeze, Rosenblum, & Joseph, 2000; Topp, Iversen, Baldry, & Maher, 2013). In the present study, being homeless during the previous 6 months was significantly associated with HIV infection among new injectors. In contrast, homeless long-term injectors presented a lower probability of being HIV positive suggesting an interaction between homelessness and length of injecting career. The worse health conditions of people infected for a longer period of time together with the higher proportion receiving a retirement pension or incapacity benefit (37.3% among HCV-positive long-term injectors) might in part explain these differences. Having had an STI was significantly associated with HIV infection among new injectors, thus highlighting the current importance of sexual transmission of HIV in this group, as reported elsewhere (Folch et al., 2013). In fact, a higher prevalence of life-time STI was expected among long-term injectors due to the accumulate risk over time. However, the similar prevalences observed among both groups are suggesting a higher incidence in the new injectors.

The association between cocaine injecting and HIV among long-term injectors had been previously described in Catalonia (Huntington et al., 2010), and it could be related to the frequency of injecting or due to increased risk behaviors among those who use cocaine, such as sharing syringes. On the other hand, frequency of injection was associated with HCV among new injectors, consistent with previous studies among this population (Judd et al., 2005).

Finally, the association between use of condoms and HIV infection among both new and long-term injectors could be explained by the implementation of measures among HIV-infected people to prevent transmission of the virus to their partners.

Approximately half of the sample of new injectors who were HIV- and/or HCV-positive did not know they were infected, thus highlighting the importance of improving early diagnosis of these infections in this group so that appropriate treatment can be provided and behavior changed. Undiagnosed infections were extremely high among the group of migrant injectors, in particular among the new injectors (71.4% for HIV and 57.4% for HCV), suggesting lower access of migrants to HIV/HCV testing. Recent observations indicate that migrants generally have limited access to health care services and harm reduction services (Rachlis et al., 2007; Harocopos, Goldsamt, Kobrak, Jost, & Clatts, 2009; Saigí et al., 2014). In this sense, it is necessary to find innovative ways of reaching this population adapted to different circumstances according to local conditions (Dietze et al., 2012), including rapid tests to detect both HIV and HCV (Drobnik, Judd, Banach, Egger, Konty, & Rude, 2011), the promotion of prevention activities through social networks, peer work (Smyrnov, Broadhead, Datsenko, & Matiyash, 2012), mobile health outreach for homeless population on the streets or in shelters or syringe dispensing machines (Jones, Pickering, Sumnall, McVeigh, J & Bellis, 2010).

This study is subject to the following limitations. First, the sample was representative of those injectors who attend Catalonian harm reduction centres, since assignment to strata was proportional to the volume of visits in each centre and to the percentage of individuals in each centre by country of birth, and since the sample selection was random within the centres. We have no information on the injectors that were not approached by these centres. Second, the validity of data could be questioned, since the prevalence of some risk behaviors may have been underestimated, even though data collectors attempted to create an anonymous atmosphere for the interviews and used simple and understandable language, however it is not obvious that this bias would be different between new and long-term injectors. Furthermore, research has demonstrated that self-reported risk behaviors can be valid and may not be strongly affected by social desirability bias (Goldstein, Friedman, Neaigus, Jose, Ildefonso, & Curtis, 1995). Caution must be taken with regard to the estimated HIV/HCV incidences because the actual date of infection is unknown. Moreover, it was not possible to know how many individuals had cleared the virus because we did not perform the determination of the viral RNA in sera. No risk factors for HCV were found among long-term injectors probably due to the limitation of current variables associated with an infection that occurred a long time ago. On the other hand, the study only collected HCV antibodies, so current infection cannot be determined. Finally, cross-sectional surveys do not allow for determination of the sequence of cause and effect which complicates interpretation of associations.

In conclusion, vulnerable populations like PWID are not homogeneous. New injectors in Catalonia, a subgroup with large proportion of migrants, present a high level of both sexual and parenteral exposure and a high incidence rate of HIV and HCV infections, therefore acting as a core group with a high basic reproductive number. Develop and strengthen early intervention initiatives to reduce the number of at risk individuals and to provide education early in injecting careers would be needed. Such interventions should be specifically focused on migrant, adapting prevention messages to their social and cultural background. Furthermore, identifying and addressing risk factors for homelessness among new injectors should be a priority. Finally, we emphasize the critical importance of improving access to HIV and HCV testing among new injectors, including outreach services to reach injectors at higher risk of these infections who may be hidden or marginalized and not in touch with traditional healthcare services.

Glossary

Harm reduction: Policies, programmes, and approaches that seek to reduce the harmful health, social, and economic consequences associated with the use of psychoactive substances

HCV: Hepatitis C virus

HIV: Human immunodeficiency virus

Incidence: The number of new cases which develop over a defined time period in a defined population at risk, divided by the number of people in that population at risk.

Prevalence rate: The number of existing cases in a defined population at a given point in time or over a defined time period, divided by the number of people in that population.

PWID: People who inject drugs

STI: Sexually transmitted Infections

Declaration of interest

The Ethics Committee of Hospital Universitari Germans Trias i Pujol approved the study. The authors report no conflicts of interest. The authors alone are responsible for the content and writing of the article.

Notes on contributors

Cinta Folch, MPH, PhD, is an epidemiologist at the Center for Epidemiological Studies on STIs, HIV and AIDS of Catalonia, Spain (CEEISCAT). She coordinates bio-behavioral HIV/STI surveillance studies among vulnerable populations since 2002 (men who have sex with men, female sex workers and intravenous drug users). She has published several national and international articles in this field, and collaborated in several European Union financed projects, such as SIALON I, SIALON II, and EMIS. Her areas of expertise include: HIV/STI Monitoring and Evaluation; Biobehavioral Surveys, Sexual Health, Vulnerable populations.

Jordi Casabona, MD, MPH, PhD (Spain), is a medical epidemiologist and the Director of the "Center for Epidemiological Studies on STIs, HIV and AIDS of Catalonia" (CEEISCAT). The Center is responsible for the regional surveillance activities on HIV and STIs and it also have an applied research component, from which he has coordinated and collaborated in several European Union financed projects, such as SIALON, SIALON II, HIV-COBATEST and Euro HIV EDAT (currently active). He is also President of the Fundació Sida i Societat and Associated Professor of Epidemiology and Preventive Medicine at the Autonomous University of Barcelona (UAB).

Albert Espelt is a Psychologist and a Statistician, and has a PhD in Public Health. He is in charge of the Drug Information System of Barcelona and works for the Public Health Agency of Barcelona (Department of Prevention and Treatment of Drug Addiction). He is an associate professor at the Universitat Autonoma de Barcelona (Department of Psychobiology and Methodology of Health Sciences (Faculty of Psychology)). He has published more than 60 papers on drug addiction and social determinants.

Xavier Majó is a Medical doctor, working since 1990 on the Programme on Substance Abuse of the Department of Health (Autonomous Government of Catalonia). From 1995 until 2014 he was in charge of policy planning and strategic development of Harm reduction programmes in Catalonia, and from 2014 he is in charge of Epidemiological surveillance of drug consumption and problems related to drug abuse. He worked for 5 years as a clinician in different drug care centres and he did a post medical degree on addictions in the National Addiction Centre of London (UK) in 1992–93 and later on, he got the specialization in Public Health (2003). He has participated in different papers and presentations, mostly on harm reduction and surveillance problems related to drug abuse.

Mercè Meroño is a Psychologist and Master in Public Health. She is a coordinator of health and social care (harm reduction) programs targeting IDUs and men, transgenders and female sex workers in Àmbit Prevenció. Other activities: Pre and Post Counselling on HIV/AIDS and other sexually transmitted diseases; Participation and Coordination of the Workshops “Empowering”, “Gender Violence”, “Transgender”; Trainer of health professionals on issues related to immigrant women and sexuality; Training and Awareness for change in attitudes and knowledge for health professionals (nurses and doctors), local service professionals (municipal councils and associations …) police professionals and the general public.

Victoria Gonzalez, MPH, PhD, is a microbiologist at the Center for Epidemiological Studies on STIs, HIV and AIDS of Catalonia, Spain (CEEISCAT). She works at the Microbiology Service of “Hospital Universitari Germans Trias i Pujol” (Badalona, Barcelona, Spain) analyzing the biological specimens collected to perform bio-behavioral HIV/STI surveillance studies among vulnerable populations. She has several national and international publications in the HIV/HCV/STIs field. Her areas of expertise include serological and molecular diagnosis of HIV, hepatitis C, hepatitis B, and STIs.

Lucas Wiessing is an epidemiologist and Principal scientist at the European Monitoring Centre for Drugs and Drug Addiction (EMCDDA) in Lisbon. He co-ordinates international projects to monitor, analyze and improve data on drug use and drug related consequences—e.g. HIV and viral hepatitis in injecting drug users—and interventions, in the European Union. Lucas is from the Netherlands, where he worked at the National Institute of Public Health and the Environment (RIVM), organizing multiple bio-behavioral HIV studies in different risk groups, in particular injecting drug users. He has published extensively and works with many relevant organizations and colleagues in his field, including key international expert and advisory groups and scientific conferences.

Joan Colom, MPH, PhD medical doctor with experience in the organization of drug addiction care network in Catalonia. He is currently the Director of the Program on Substance Abuse of the Department of Health (Autonomous Government of Catalonia). Diploma in Public Health (National Health School), Masters in Directing ESADE professor Master Drug Addiction at the University of Barcelona, author of articles and national and international publications in the field of drug addiction, author lectures and papers in meetings and conferences of national and international advisor to the World Health Organization, a member of 20 scientific societies and committees, “White Cross” of Merit of the National Plan on Drugs, “Queen Sofia” Prize for drug prevention Prize for the best Research (Diario Médico), Chairman of the Public Management Club of the College of Business Administration, founder INEBRIA.

M. Teresa Brugal, MD, PhD, MPH (Barcelona Autonomous University), is senior researcher on epidemiology of drug abuse and related diseases. She works in the Public Health Agency of Barcelona as Chief of department of Prevention and Treatment of Addictions. Her works consists in planning and evaluate drug plans, policies in the drug field, as well as to evaluate the efficacy of interventions and treatments implemented by the different Spanish health administrations. In recent years she has developed several applied research studies, mainly in the field of drug addiction and associated diseases (psychiatric and infectious) and in evaluating treatments (methadone, naloxone, etc.). She has also conducted cohort studies among drug users, to analyze the evolution of consumer habits, practices and associated mortality risk. These investigations have resulted in over 100 scientific papers published in international medical literature.

REDAN Study Group is comprised of Jordi Casabona, Cinta Folch, Anna Esteve, Alexandra Montoliu, and Rafa Muñoz (CEEISCAT); Victoria Gonzalez and Vicenç Ausina (Servei Microbiologia HUGTiP); M. Teresa Brugal, Albert Espelt, and Carmen Vecino (Agència de Salut Pública de Barcelona); Xavier Majó and Joan Colom (Subdirecció General de Drogodependències); and Mercè Meroño and Anna Altabas (Àmbit Prevenció).

References

Bailey, S. L., Ouellet, L. J., Mackesy-Amiti, M. E., Golub, E. T., Hagan, H., Hudson, S. M., … DUIT Study Team. (2007). Perceived risk, peer influences, and injection partner type predict receptive syringe sharing among young adult injection drug users in five U.S. cities. *Drug and Alcohol Dependance*, *91*(1), S18–29.

Barrio, G., Bravo, M. J., Brugal, M. T., Diez, M., Regidor, E., Belza, M. J., … Itinere Working Group. (2012). Harm reduction interventions for drug injectors or heroin users in Spain: Expanding coverage as the storm abates. *Addiction*, *107*(6), 1111–1122.

Barros, A. J. D., & Hirakata, V. N. (2003). Alternatives for logistic regression in cross-sectional studies: An empirical comparison of models that directly estimate the prevalence ratio. *BMC Medical Research Methodology*, *3*, 21.

Becker Buxton, M., Vlahov, D., Strathdee, S. A., Des Jarlais, D. C., Morse, E. V., Ouellet, L., … Garfein, R. S. (2004). Association between injection practices and duration of injection among recently initiated injection drug users. *Drug and Alcohol Dependence*, *75*(2), 177–183.

Bravo, M. J., Vallejo, F., Barrio, G., Brugal, M. T., Molist, G., Pulido, J., … Itinere Working Group. (2012). HCV seroconversion among never-injecting heroin users at baseline: No predictors identified other than starting injection. *The International Journal on Drug Policy*, *23*(5), 415–419.

Centre d'Estudis Epidemiològics sobre les Infeccions de Transmissió Sexual i Sida de Catalunya (CEEISCAT). (2013). *Integrated surveillance of HIV/AIDS/STI Catalonia (SIVES 2012)*. (Technical Document 21). Barcelona: Generalitat de Catalunya, Agència de Salut Pública de Catalunya.Retrieved from http://www.ceeiscat.cat/documents/SIVES_2012.pdf

Chohan, B. H., Lavreys, L., Mandaliya, K. N., Kreiss, J. K., Bwayo, J. J., Ndinya-Achola, J. O., & Martin H. L., Jr. (2001). Validation of a modified commercial enzyme-linked immunoassay for detection of human immunodeficiency virus type 1 immunoglobulin G antibodies in saliva. *Clinical and Diagnostic Laboratory Immunology*, *8*(2), 346–348.

De la Fuente, L., Bravo, M. J., Toro, C., Brugal, M. T., Barrio, G., Soriano, V., … Itinere Working Group. (2006). Injecting and HIV prevalence among young heroin users in three Spanish cities and their association with the delayed implementation of harm reduction programmes. *Journal of Epidemiology and Community Health*, *60*(6), 537–542.

Dietze, P., Winter, R., Pedrana, A., Leicht, A., Majó, X., & Brugal, M.T. (2012). Mobile safe injecting facilities in barcelona and Berlin. *International Journal of Drug Policy*, *23*, 257–260.

Drobnik, A., Judd, C., Banach, D., Egger, J., Konty, K., & Rude, E. (2011). Public health implications of rapid hepatitis C screening with an oral swab for community-based organizations serving high-risk populations. *American Journal of Public Health*, *101*(11), 2151–2155.

European Monitoring Center for Drugs and Drug Addiction (EMCDDA). (2012). *Annual Report 2012: The State of the Drugs Problem in Europe*. Luxembourg: Publications Office of the European Union.

Folch, C., Casabona, J., Espelt, A., Majo, X., Merono, M., Gonzalez, V., … REDAN Study Group. (2013). Gender differences in HIV risk behaviours among intravenous drug users in Catalonia, Spain. *Gaceta Sanitaria/S.E.S.P.A.S*, *27*(4), 338–343.

Freixa, P., & Merono, M. (2011). *Estudi qualitatiu sobre usuaris de drogues per via parenteral provinents d'Europa de l'Est a Catalunya*. Barcelona, Spain: Àmbit Prevenció.

Friedman, S.R., Rossi, D., & Braine, N. (2009). Theorizing "Big Events" as a potential risk environment for drug use, drug-related harm and HIV epidemic outbreaks. *International Journal of Drug Policy*, *20*, 283–91.

Garten, R. J., Lai, S., Zhang, J., Liu, W., Chen, J., Vlahov, D., & Yu, X. F. (2004). Rapid transmission of hepatitis C virus among young injecting heroin users in Southern China. *International Journal of Epidemiology*, *33*(1), 182–188.

Goldstein, M. F., Friedman, S. R., Neaigus, A., Jose, B., Ildefonso, G., & Curtis, R. (1995). Self-reports of HIV risk behavior by injecting drug users: Are they reliable? *Addiction*, *90*(8), 1097–1104.

Hacker, M. A., Friedman, S. R., Telles, P. R., Teixeira, S. L., Bongertz, V., Morgado, M. G., & Bastos, F. I. (2005). The role of "long-term" and "new" injectors in a declining HIV/AIDS epidemic in Rio de Janeiro, Brazil. *Substance Use & Misuse*, *40*(1), 99–123.

Hagan, H., Pouget, E. R., Des Jarlais, D. C., & Lelutiu-Weinberger, C. (2008). Meta-regression of hepatitis C virus infection in relation to time since onset of illicit drug injection: The influence of time and place. *American Journal of Epidemiology*, *168*(10), 1099–1109.

Hagan, H., Pouget, E. R., Williams, I. T., Garfein, R. L., Strathdee, S. A., Hudson, S. M., … Ouellet, L. J. (2010). Attribution of hepatitis C virus seroconversion risk in young injection drug users in 5 US cities. *Journal of Infectious Diseases, 201*(3), 378–385.

Harocopos, A., Goldsamt, L. A., Kobrak, P., Jost, J. J., & Clatts, M. C. (2009). New injectors and the social context of injection initiation. *The International Journal on Drug Policy*, *20*, 317–323.

Huntington, S., Folch, C., Gonzalez, V., Meroño, M., Ncube, F., & Casabona, J. (2010). Prevalence of human immunodeficiency virus and hepatitis C virus, and associated factors among injecting drug users in Catalonia. *Enfermedades Infecciosas y Microbiologia Clinica*, *28*(4), 236–238.

Jones, L., Pickering, L., Sumnall, H., McVeigh, J., & Bellis, M. A. (2010). Optimal provision of needle and syringe programmes for injecting drug users: A systematic review. *International Journal of Drug Policy*, *21*(5), 335–342.

Judd, A., Hickman, M., Jones, S., McDonald, T., Parry, J. V, Stimson, G. V, & Hall, A. J. (2005). Incidence of hepatitis C virus and HIV among new injecting drug users in London: Prospective cohort study. *British Medical Journal*, *330*(7481), 24–25.

Judd, A., Parry, J., Hickman, M., McDonald, T., Jordan, L., Lewis, K., … Nelson, M. (2003). Evaluation of a modified commercial assay in detecting antibody to hepatitis C virus in oral fluids and dried blood spots. *Journal of Medical Virology*, *71*(1), 49–55.

Magura, S., Nwakeze, P. C., Rosenblum, A., & Joseph, H. (2000). Substance misuse and related infectious diseases in a soup kitchen population. *Substance Use & Misuse*, *35*(4), 551–583.

Maher, L., Li, J., Jalaludin, B., Chant, K. G., & Kaldor, J. M. (2007). High hepatitis C incidence in new injecting drug users: A policy failure? *Australian and New Zealand Journal of Public Health*, *31*(1), 30–35.

Miller, C. L., Johnston, C., Spittal, P. M., Li, K., Laliberte, N., Montaner, J. S., & Schechter, M. T. (2002). Opportunities for prevention: Hepatitis C prevalence and incidence in a cohort of young injection drug users. *Hepatology*, *36*(3), 737–742.

Neaigus, A., Gyarmathy, V. A., Miller, M., Frajzyngier, V., Zhao, M., Friedman, S. R., & Des Jarlais, D. C. (2007). Injecting and sexual risk correlates of HBV and HCV seroprevalence among new drug injectors. *Drug and Alcohol Dependence*, *89*(2–3), 234–243.

Paraskevis, D., Nikolopoulos, G., Fotiou, A., Tsiara, C., Paraskeva, D., Sypsa, V., … Hatzakis, A. (2013). Economic recession and emergence of an HIV-1 outbreak among drug injectors in athens metropolitan area: A longitudinal study. *PloS One*, *8*, e78941. doi:10.1371/journal.pone.0078941

Pouget, E. R., Hagan, H., & Des Jarlais, D. C. (2012). Meta-analysis of hepatitis C seroconversion in relation to shared syringes and drug preparation equipment. *Addiction*, *107*(6), 1057–1065.

Quan, V. M., Go, V. F., Nam le, V., Bergenstrom, A., Thuoc, N. P., Zenilman, J., … Celentano, D. D. (2009). Risks for HIV, HBV, and HCV infections among male injection drug users in northern Vietnam: A case-control study. *AIDS Care*, *21*(1), 7–16.

Rachlis, B., Brouwer, K. C., Mills, E. J., Hayes, M., Kerr, T., & Hogg, R. S. (2007). Migration and transmission of blood-borne infections among injection drug users: Understanding the epidemiologic bridge. *Drug and Alcohol Dependence*, *90*(2–3), 107–119.

Reyes, J. C., Robles, R. R., Colón, H. M., Matos, T. D., Finlinson, H. A., Marrero, C. A., & Shepard, E. W. (2005). Homelessness and HIV risk behaviors among drug injectors in Puerto Rico. *Journal of Urban Health*, *82*, 446–455.

Saigí, N., Espelt, A., Folch, C., Sarasa-Renedo, A., Castellano, Y., Majó, X., Meroño, M., … REDAN Study Group. (2014). Differences in illegal drug consumption between native and immigrants in a large sample of injected drug users in Catalonia (Spain). *Adicciones*, *26*, 69–76.

Smyrnov, P., Broadhead, R. S., Datsenko, O., & Matiyash, O. (2012). Rejuvenating harm reduction projects for injection drug users: Ukraine's nationwide introduction of peer-driven interventions. *International Journal of Drug Policy*, *23*, 141–147.

Strathdee, S.A., Lozada, R., Pollini, R.A., Brouner, K.C., Mantsios, A., Abramovitz, D. A., Rhodes, … Patterson, T.L. (2008). Individual, social, and environmental influences associated with HIV infection among injection drug users in Tijuana, Mexico. *Journal of Acquired Immune Deficiency Syndromes*, *47*, 369–376.

Sysa, V., Paraskevis, D., Malliori, M., Nikopoulos, G.K., Panopoulos, A., Kantzanou, M., Katsoulidou, A., … Hatzakis, A. (2015). Homelessness and other risk factors for HIV infection in the current outbreak among injection drug users in Athens, Greece. *American Journal of Public Health*, *1*, 196–204.

Thiede, H., Hagan, H., Campbell, J. V, Strathdee, S. A., Bailey, S. L., Hudson, S. M., … Team, D. S. (2007). Prevalence and correlates of indirect sharing practices among young adult injection drug users in five U.S. cities. *Drug and Alcohol Dependence*, *91*(Suppl 1), S39–47.

Topp, L., Iversen, J., Baldry, E., & Maher, L. (2013). Housing instability among people who inject drugs: Results from the Australian needle and syringe program survey. *Journal of Urban Health*, *90*(4), 699–716.

Tordable Merino, I., Sánchez Sánchez, A., Santos Sanz, S., García Vicario, M., & Redondo Martín, S. (2010).Trends in drug consumption among immigrants between 2004 and 2008. *Gaceta Sanitaria/S.E.S.P.A.S, 24*(3), 200–203.

Uusküla, A., Kals, M., Rajaleid, K., Abel, K., Talu, A., Rüütel, K., … Des Jarlais, D. (2008). High-prevalence and high-estimated incidence of HIV infection among new injecting drug users in Estonia: Need for large scale prevention programs. *Journal of Public Health*, *30*(2), 119–125.

Wiessing, L., Likatavicius, G., Hedrich, D., Guarita, B., van de Laar, M. J., & Vicente, J. (2011). Trends in HIV and hepatitis C virus infections among injecting drug users in Europe, 2005 to 2010. *Euro surveillance*, *16*(48), 20031. Retrieved from http://www.eurosurveillance.org/ViewArticle.aspx?ArticleId=20031

World Health Organisation (WHO). (1994). *Multi-City Study on Drug Injecting and Risk of HIV Infection. Programme on Substance Abuse—Final Report*. WHO/PSA/94.4. Geneva, Switzerland: Author.